

Medical SAM Adapter++: Adaptive Normalization with Domain Interpretation for Better Adaptation

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Abstract—Vision Transformer (ViT) models have spearheaded the computer vision community for various image-related tasks, specifically showing significant results in interactive segmentation tasks using Segment Anything Model (SAM). Adapters can be incorporated into SAM for adaptation to the different domains, whilst significantly reducing training costs compared to the fine-tuning of the entire model. This was especially true for datasets from the medical domain, where it presented results superior to state-of-the-art models. However, the simple incorporation of adapters fails to leverage the statistical differences between the probability distribution of data from various domains. This paper addresses this issue by introducing Adaptive Layer Normalization into SAM, which has more theoretical background as a solution to domain differences than sequential adapters. Furthermore, this paper also proposes a novel method, domain interpreter, for determining the weights to Adaptive Layer Normalization. Focusing on the Osteoarthritis (OAI) Dataset, the proposed method shows improved results in IoU and DICE metric scores, showing that the distinct domain distributions are the cause of inaccuracies of baseline models, and that the models can quickly be adapted to different domains by adjusting to such data distributions.

Index Terms—medical AI, segmentation, adapters

I. INTRODUCTION

In recent years, the field of image segmentation has been revolutionized by deep learning models such as Meta’s Segment Anything Model (SAM) [7]. SAM has set a new benchmark with its impressive performance and versatility in various applications. However, like many general-purpose models, SAM was initially trained for broad applicability without specific domain features trained into its architecture. This limitation is particularly apparent when applying the model to specialized domains such as medical imaging, where task-specific features are critical but often require fine-tuning of the model to adapt and learn these unique characteristics.

Focusing on the medical domain, an example of such characteristics is present in Magnetic Resonance Images (MRI) of the knee, where different significant musculoskeletal regions for diagnosing Osteoarthritis (OA) are separated only by a low contrast intensity shift. Simultaneously, insignificant muscular regions found in the images have a more pronounced, high contrast intensity shift that is falsely recognizable as a potential

boundary by the baseline SAM architecture. Such features inherent to the image cause medical image segmentation to be more challenging.

A solution for segmentation inaccuracies in the aforementioned example is to fine-tune the model with data from the unseen domain [3]. However, fine-tuning a model requires large computational costs, a limitation inherent to transformer architectures. A preceding study [2] addressed this limitation by integrating adapters—a concept derived from transformer networks for context learning—into SAM’s architecture. Using this technique, the model successfully presented results superior to state-of-the-art (SOTA) models. However, the presented methods failed to fully utilize the derivable features of an image, providing potential for improvements to the proposed model.

Motivated by this shortcoming, we propose an innovative solution: the incorporation of Adaptive Layer Norm (AdaLN) [4] into the adapter architecture for SAM. This modification is designed to adapt more effectively to different domains with varying conditions and characteristics, ensuring better segmentation outcomes even in complex medical imaging scenarios where traditional methods may struggle. By doing so, we aim to enhance SAM’s ability to handle domain-specific nuances, thereby improving its applicability across a broader spectrum of tasks and domains.

In the following sections, we will detail our proposed method, discuss the experimental setup, and present the results that demonstrate the superiority of AdaLN over traditional MLP adapters in adapting SAM for improved performance on specialized datasets, specifically, the OAI knee dataset. We believe these contributions not only advance the state-of-the-art in image segmentation but also provide a more robust foundation for future research into model adaptability and specificity. The main contributions of the paper include:

- 1) Incorporation of Adaptive Layer Normalization (AdaLN) for better domain adaptation in Med-SA.
- 2) Novel context-based Domain Interpreter for incorporation of global features.
- 3) Improved Med-SA model for segmentation in Medical Images.

II. METHODOLOGY

A. Preliminaries: Med-SA Architecture

To begin, we give a brief overview of the Med-SA architecture. Med-SA provided three main contributions to the original SAM model. The contributions consisted of the incorporation of adapters into the SAM encoder module, a SD-Trans module capable of finding depth-wise patch relationships for 3D inputs, and a Hyper-Prompting Adapter for the user-input prompt. For this paper, we focused only on the 2-dimensional segmentation capabilities, leaving the SD-Trans module unused due to its computational requirements. The adapters within the SAM encoder module consisted of a bottleneck Multi-Layer Perceptron (MLP), with a GELU activation in the center. Two adapters were placed: one after the self-attention layer, and the other parallel to the layer normalization layer. Furthermore, a Hyper-Prompting Adapter was applied to the SAM Decoder module, where hyper prompts derived from the input prompt embedding were utilized as weights to the downsized vectors in the adapter module, before the activation layer. These changes allowed the SAM model to be fine-tuned for specific tasks while keeping its trainable parameter count low.

B. Adaptive Layer Normalization

In this section, we introduce the practicality and the applicability of Adaptive Layer Normalizations (AdaLN) to Med-SA. As shown in Fig. 2, in the original Med-SA encoder, the embeddings pass through a linear normalization layer, where the embeddings are normalized to have a mean and standard deviation of zero and one. The specific calculations are:

$$\begin{aligned}\mu_i &= \frac{1}{K} \sum_{k=1}^K \mathbf{x}_{i,k} \\ \sigma_i^2 &= \frac{1}{K} \sum_{k=1}^K (\mathbf{x}_{i,k} - \mu_i)^2\end{aligned}\quad (1)$$

where μ_i and σ_i^2 represent the mean and variance of the data in the i^{th} layer, respectively.

These values are then applied to the input vector $\mathbf{x}$ for normalization as follows:

$$\tilde{\mathbf{x}}_{i,k} = \frac{\mathbf{x}_{i,k} - \mu_i}{\sqrt{\sigma_i^2 + \epsilon}} \quad (2)$$

where ϵ is used for numerical stability.

AdaLN uses the same concept from Med-SA to freeze the original parameters from SAM, but instead of adding sequential adapters, added learnable parameters γ and β to shift and scale the embedded vectors. The specific transformation applied to the normalized feature is as follows:

$$\mathbf{o}_i = \gamma \tilde{\mathbf{x}}_i + \beta \quad (3)$$

As is shown in different studies [6], this shifting and scaling adhere to the theory that the additional process aids the model in capturing the statistical information of the target domain from a source domain. The distinctiveness of statistical

information between data distributions is also demonstrated in Fig. 1.

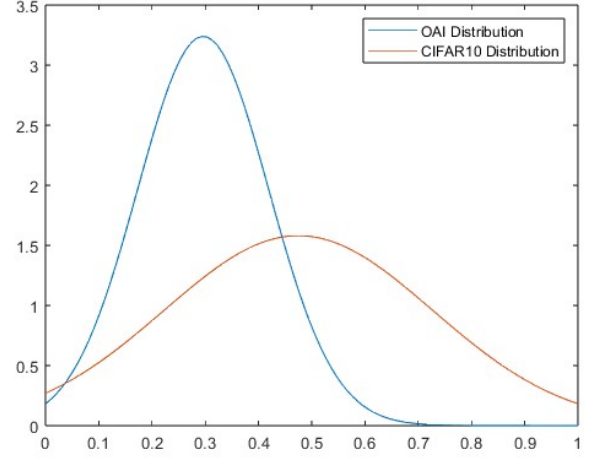


Fig. 1. Normal distributions depicted from the means and variances of CIFAR10 and OAI dataset

Assuming that the target mask output dataset can be represented by a probability distribution, with given conditions input image $\mathbf{x}$ and point condition $\mathbf{c}$, the loss function can be represented as the following:

$$\mathcal{L}_\theta = ||P(m_{gt}) - P(m_{pred}|\mathbf{x}, \mathbf{c})|| \quad (4)$$

where $\mathcal{L}$ refers to the L1 loss function, θ refers to the model weights, $P(m_{gt})$ and $P(m_{pred}|\mathbf{x}, \mathbf{c})$ refers to the ground truth mask probability distribution and the predicted mask probability distribution based on the given image inputs $\mathbf{x}$ and point conditions $\mathbf{c}$.

Calculating for the minima of the loss function, we find that the following equation should hold true:

$$\text{argmin}(\mathcal{L}_\theta) = P(m_{pred}|\mathbf{x}, \mathbf{c}) \sim P(m_{gt}) \quad (5)$$

where the distribution of the predicted outputs should match the distribution of the ground truth segmentation. In other words, the probability distribution of the output of the model can be adjusted through the parameter updates in AdaLN for the distribution to match the desired target domain.

Aiming to improve the results obtained from Med-SA, this paper inserts the AdaLN layer as an addition to the sequential adapter used in Med-SA for better domain adaptation.

C. Domain Interpreter

In this section, we explain how further guidance can be given to learn γ and β . As was discussed in the previous section, difficulties in segmenting images from different domains can be traced to the uniqueness of their data distributions. These data distributions are innate to the input images and can provide further context for the model. Therefore, we propose Domain Interpreter (DI), a module that receives contexts from the input image and outputs the parameters γ and β in AdaLN.

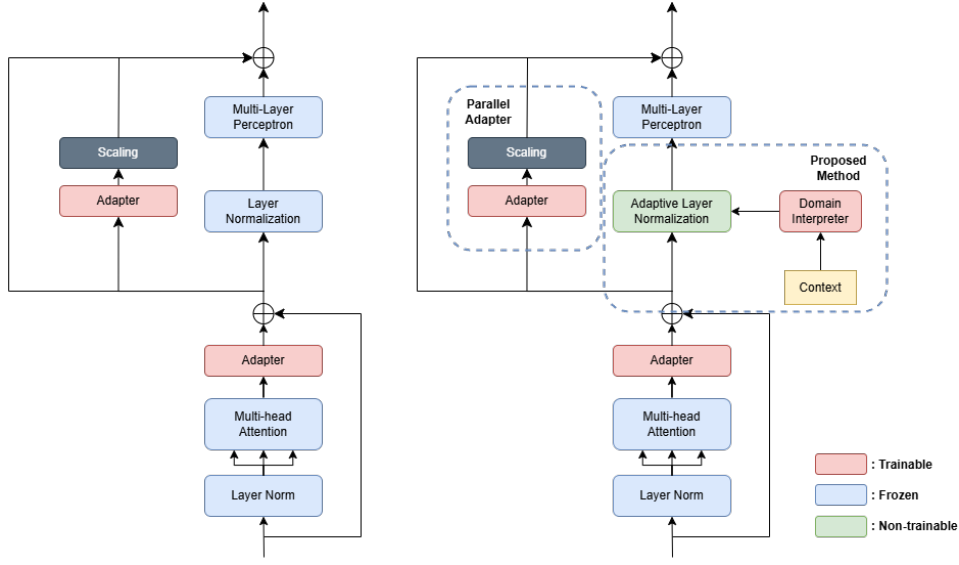


Fig. 2. Direct comparison between Med-SA encoder on the left and the proposed method on the right.

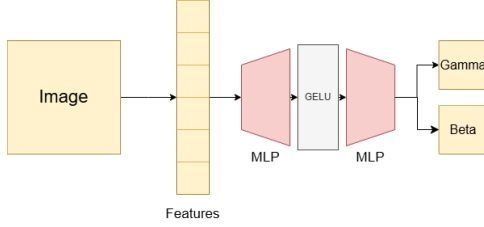


Fig. 3. Detailed architecture of DI module.

As shown in Fig. 2, the DI receives an input context from the given image. The demonstration in Fig. 1 shows that the characteristics of the dataset are distinct in their distribution graph, which can be estimated to a normal distribution, $X \sim \mathcal{N}(\mu, \sigma^2)$, unique to each dataset. Therefore, the means and variances of the pixel intensity values for each channel were used as the input features to the DI module. The calculation for these features is as follows:

$$\begin{aligned} \mu_c &= \frac{1}{N \times M} \sum_{i=1}^N \sum_{j=1}^M x_{i,j} \\ \sigma_c^2 &= \frac{1}{N \times M} \sum_{i=1}^N \sum_{j=1}^M (x_{i,j} - \mu_c)^2 \end{aligned} \quad (6)$$

where c represents the channel, N and M represent the size of the image, and x represents the specific pixel value at position (i, j) .

The extracted contexts are then processed by two MLP layers, with an activation function in between them. The expansion in the MLP layers aims to capture specific human undefined features from the given context. The module then shrinks the output to two values, γ and β . The detailed architecture can be found in Fig. 3.

D. Dataset

For this paper, we focused on two datasets from the medical domain, the Osteoarthritis Initiative (OAI) dataset [1], which is publicly available (<https://nda.nih.gov/oai>), and the REFUGE dataset, which was provided by [8].

1) *OAI Dataset*: The OAI dataset consists of knee MRI images with cartilage and meniscus segmentation maps. The significance of knee MRI segmentation exists in knee osteoarthritis diagnosis, which has been a research topic of interest due to the increase in global occurrences, accounting for more than 56% of the total number of osteoarthritis occurrences [1]. Specifically for segmentation, the dataset consists of segmentation maps from 88 anonymous subjects, whose MRI's consist of 124 annotated slices. This totals to 24,283 annotated MRI slices of the knee. 6 distinct labels are available for the dataset, each labeling the different cartilage and menisci. As the goal of an interactive segmentation model is not to predict all possible segmentation maps in an image, but to predict a segmentation map specific to a given input, these distinct labels were used to simulate single point prompts to the model, similar to the methods used in [2].

2) *REFUGE Dataset*: The REFUGE dataset comprises retinal fundus images annotated for optic disc and optic cup segmentation, which are critical for glaucoma detection and assessment. Glaucoma is a major cause of irreversible blindness worldwide, and early diagnosis through accurate optic disc and cup segmentation has been a central focus in ophthalmic research [8]. The dataset includes images from multiple sources with a standardized resolution of 2124x2056 pixels, which were reshaped to 512x512 pixels for computational purposes, and each image is accompanied by pixel-wise annotations provided by clinical experts. A total of 1,200 fundus images are available, split into training, validation, and test sets, with corresponding ground truth segmentation

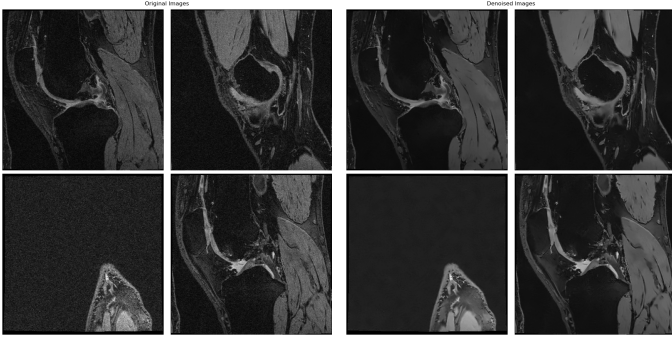


Fig. 4. Original and denoised OAI dataset. Left is the noisy dataset, whilst the right is the clean dataset. Noisy images in each corner correspond to its clean counterpart in its corner.

masks for the optic disc and optic cup. Individual regions were selected as target masks, simulating user-guided segmentation prompts in a manner consistent with the approach for the OAI Dataset.

Applied only to the OAI dataset, two different versions of the dataset were used as training for the model. The first one consisted of the original images, where Gaussian noises were consistently found in darker regions of the images. The second dataset consisted of preprocessed denoised images, where potential loss of information occurred. The denoising step was processed by the fast non-local means denoising algorithm, with a denoising strength of 4, template window size of 7, and search window size of 21. Images of the original and denoised dataset can be found in Fig. 4.

III. RESULTS AND DISCUSSION

A. Experiment Setup

To evaluate the effectiveness of incorporating Adaptive Layer Normalization (AdaLN) into Med-SA, we conducted three experimental setups using the official ViT-H weights from the Segment Anything Model (SAM) repository. The experiments were designed as follows:

- 1) **Baseline Med-SA:** The original Med-SA architecture with sequential MLP adapters.
- 2) **Med-SA with Denoising:** The original Med-SA architecture with the denoised dataset.
- 3) **Med-SA++ with Denoising:** The full proposed model incorporating both AdaLN and DI with the denoised dataset.

All models were trained using the Adam optimizer on three NVIDIA V100 GPUs (32GB VRAM each), with a batch size of 4 and a learning rate of 1×10^{-4} . Evaluation was conducted using the Intersection over Union (IoU) and Dice Coefficient (DICE), two standard metrics for image segmentation quality.

B. Qualitative Results

To further illustrate the effectiveness of our proposed approach, Figure 5 presents qualitative comparisons between segmentation outputs generated by our full model (Med-SA++), the baseline Med-SA, and the ground truth annotations.

Each row displays a sample from the OAI dataset, including the input MRI image, segmentation predictions from our model and Med-SA, and the corresponding ground truth mask. As shown in the first example, the baseline Med-SA struggles to produce accurate and coherent segmentation maps, resulting in fragmented and noisy outputs. In contrast, our model generates predictions that more closely resemble the ground truth, capturing both the shape and location of the target structure with greater precision.

The second example demonstrates similar trends. While Med-SA produces partial or misaligned segmentation maps, our method yields clean and structurally consistent predictions, especially in complex curved regions. This improvement highlights the combined benefit of AdaLN and the Domain Interpreter, which together enhance domain adaptation and enable the model to better capture medical-specific anatomical structures.

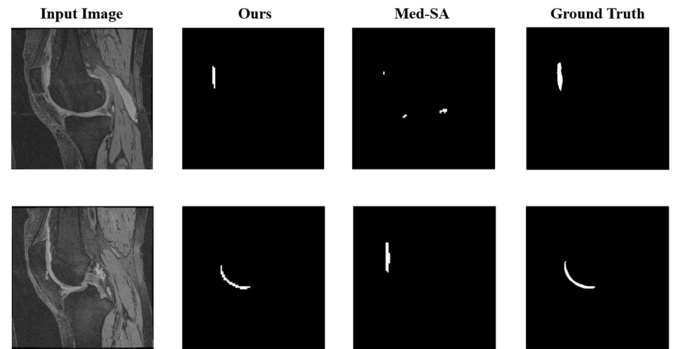


Fig. 5. Qualitative comparison of segmentation results on the OAI dataset. From left to right: input image, output from our model (Med-SA++), output from Med-SA (baseline), and the ground truth mask. Our method produces cleaner and more accurate segmentation maps that better align with the ground truth.

These qualitative results reinforce the findings of our ablation study, further demonstrating that our domain-adaptive architecture significantly improves segmentation quality over the baseline.

C. Quantitative Results

Table I presents a comparative analysis of segmentation performance between the baseline Med-SA and the proposed Med-SA++ models across the OAI and REFUGE datasets. For the OAI dataset, results are reported on the denoised version, while for the REFUGE dataset, evaluation is provided separately for the optic cup and optic disc regions.

On the denoised OAI dataset, the Med-SA++ model achieves an IoU of 58.71% and a DICE score of 71.05%, compared to the baseline (Denoised + Med-SA), which records an IoU of 58.20% and a DICE score of 70.49%. This reflects a consistent improvement of 0.9% in IoU and 0.8% in DICE scores, which exhibits the DI's segmentation performance even under reduced-noise imaging conditions. This increase in segmentation performance can also be observed on the REFUGE dataset, where Med-SA++ demonstrated relevant results. For the optic cup segmentation, it achieves an IoU

TABLE I
COMPARATIVE RESULTS OF MED-SA++ WITH BASELINE MODEL

Models	OAI		REFUGE Cup		REFUGE Disc	
	IoU $\uparrow$	DICE $\uparrow$	IoU $\uparrow$	DICE $\uparrow$	IoU $\uparrow$	DICE $\uparrow$
Med-SA	51.04%	62.54%	56.02%	69.90%	79.16%	88.03%
Denoised + Med-SA	58.20%	70.49%	-	-	-	-
Denoised + Med-SA++	58.71%	71.05%	57.00%	70.81%	80.80%	89.05%

of 57.00% and a DICE score of 70.81%, surpassing Med-SA’s performance of 56.02% IoU and 69.90% DICE. For the optic disc, Med-SA++ reaches an IoU of 80.80% and DICE of 89.05%, also improving upon Med-SA’s IoU of 79.16% and DICE of 88.03%.

D. Ablation Study

To better understand the impact of the proposed modifications, we performed an ablation study by evaluating the performance of different configurations of our model. Specifically, we evaluated three variants:

- 1) Baseline Med-SA: This model uses the original SAM architecture with sequential MLP adapters as proposed in prior work.
- 2) Med-SA + AdaLN: This configuration adds the AdaLN module to the Baseline Med-SA, allowing for domain-specific adjustments without the Domain Interpreter.
- 3) Med-SA + AdaLN + DI: This is the full model, incorporating both AdaLN and the DI module, as proposed in this paper.

Table II shows the performance of each configuration using the noisy version of the OAI dataset, evaluating IoU and DICE scores. The results reveal several key insights.

The baseline Med-SA achieves an IoU of 51.04% and a DICE score of 62.54%, establishing a solid foundation for general medical segmentation tasks. Adding AdaLN leads to a moderate boost in performance, increasing IoU to 51.42% and DICE to 64.25%. This indicates that AdaLN helps the model adapt better to domain-specific feature distributions by learning instance-aware normalization parameters, though its impact remains relatively incremental.

TABLE II
ABLATION STUDY OF ADA LN AND DI

Models	IoU $\uparrow$	DICE $\uparrow$
Med-SA	51.04%	62.54%
Med-SA+AdaLN	51.42	64.25
Med-SA+AdaLN+DI	53.61%	66.33%

The full model, Med-SA + AdaLN + DI, achieves the best performance with an IoU of 53.61% and a DICE of 66.33%, improving over the baseline by +2.57% IoU and +3.79% DICE. These gains highlight the significance of the Domain Interpreter in extracting and applying global domain-level knowledge to guide AdaLN, thereby enabling more robust domain adaptation and accurate segmentation.

IV. CONCLUSION AND FUTURE WORK

In this paper, we proposed the use of AdaLN and a novel DI module to adapt SAM to different dataset distributions for better segmentation results. In our results, we have proven that our proposed method yields better results than the baseline adapter model, signifying that our hypothesis of domain adaptation depending on data distributions holds true. Furthermore, the ablation results demonstrate that while AdaLN introduces import adaptability, it is the DI module that increases the performance of the model. This confirms the effectiveness of our Med-SA++ architecture for domain-specific medical image segmentation.

However, our work still has the potential to present better results in the IoU and DICE scores, as the accuracies were not on par with other Convolutional Neural Network (CNN)-based models. Future work will be performed to further improve these metrics.

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